

HDOC - 34

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Question

Write a C program to perform the following menu-driven operations using separate functions:

1. Read three variable-length 2D arrays and verify the left distributive property: $A*(B+C) = (A*B) + (A*C)$.
Dimensions: A (m x n), B (n x p), C (n x p).
2. Read three variable-length 2D arrays and verify the right distributive property: $(A+B)*C = (A*C) + (B*C)$.
Dimensions: A (m x n), B (m x n), C (n x p).
3. Exit

Note: Display both resulting (m x p) matrices and state whether the selected distributive property is verified.

Program

```
#include <stdio.h>

void readM(int r,int c,int a[r][c]) {
    int i,j;
    for(i=0;i<r;i++)
        for(j=0;j<c;j++) scanf("%d",&a[i][j]);
}

void printM(int r,int c,int a[r][c]) {
    int i,j;
    for(i=0;i<r;i++) {
        for(j=0;j<c;j++) printf("%d ",a[i][j]);
        printf("\n");
    }
}

void add(int r,int c,int a[r][c],int b[r][c],int s[r][c]) {
    int i,j;
    for(i=0;i<r;i++)
        for(j=0;j<c;j++) s[i][j]=a[i][j]+b[i][j];
}

void multiply(int r1,int c1,int c2,int a[r1][c1],
              int b[c1][c2],int p[r1][c2]) {
    int i,j,k;
    for(i=0;i<r1;i++)
        for(j=0;j<c2;j++) {
            p[i][j]=0;
            for(k=0;k<c1;k++) p[i][j]+=a[i][k]*b[k][j];
        }
}

int same(int r,int c,int a[r][c],int b[r][c]) {
    int i,j;
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            if(a[i][j]!=b[i][j]) return 0;
    return 1;
}

int main() {
    int choice;
    do {
        printf("\n1. Left distributive property\n");
        printf("\n2. Right distributive property\n");
        printf("\n3. Exit\nEnter your choice: ");
        scanf("%d",&choice);

        if(choice==1) {
            int m,n,p;
            printf("Enter m n p: "); scanf("%d%d%d",&m,&n,&p);
            int A[m][n],B[n][p],C[n][p],BC[n][p];
            int L[m][p],AB[m][p],AC[m][p],R[m][p];

            printf("Enter A:\n"); readM(m,n,A);
            printf("Enter B:\n"); readM(n,p,B);
            printf("Enter C:\n"); readM(n,p,C);

            add(n,p,B,C,BC);
            multiply(m,n,p,A,BC,L);
            multiply(m,n,p,A,B,AB);
            multiply(m,n,p,A,C,AC);
            add(m,p,AB,AC,R);

            printf("A*(B+C):\n"); printM(m,p,L);
            printf("(A*B)+(A*C):\n"); printM(m,p,R);
            if(same(m,p,L,R))
                printf("Left distributive property verified.\n");
        }
    } while(choice!=3);
}
```

```

        else printf("Property not verified.\n");
    }
    else if(choice==2) {
        int m,n,p;
        printf("Enter m n p: "); scanf("%d%d%d",&m,&n,&p);
        int A[m][n],B[m][n],C[n][p],AB[m][n];
        int L[m][p],AC[m][p],BC[m][p],R[m][p];

        printf("Enter A:\n"); readM(m,n,A);
        printf("Enter B:\n"); readM(m,n,B);
        printf("Enter C:\n"); readM(n,p,C);

        add(m,n,A,B,AB);
        multiply(m,n,p,AB,C,L);
        multiply(m,n,p,A,C,AC);
        multiply(m,n,p,B,C,BC);
        add(m,p,AC,BC,R);

        printf("(A+B)*C:\n"); printM(m,p,L);
        printf("(A*C)+(B*C):\n"); printM(m,p,R);
        if(same(m,p,L,R))
            printf("Right distributive property verified.\n");
        else printf("Property not verified.\n");
    }
    else if(choice==3) printf("Exiting program...\n");
    else printf("Invalid choice\n");
} while(choice!=3);
return 0;
}

```

Sample Output

```

1. Left distributive property
2. Right distributive property
3. Exit
Enter your choice: 1
Enter m n p: 2 2 2
Enter A:
1 2
3 4
Enter B:
1 0
0 1
Enter C:
2 1
1 2
A*(B+C):
5 6
11 12
(A*B)+(A*C):
5 6
11 12
Left distributive property verified.

```

```

Enter your choice: 2
Enter m n p: 2 2 2
Enter A:
1 2
3 4
Enter B:
5 6
7 8
Enter C:
1 0
0 1
(A+B)*C:
6 8
10 12
(A*C)+(B*C):
6 8
10 12
Right distributive property verified.

```

```

Enter your choice: 3
Exiting program...

```

Result: The program successfully verifies the left and right distributive properties of matrix multiplication over matrix addition using separate functions and variable-length 2D arrays.